

ASSIGNMENT-2

1. Epoch, Batch Size, and Stratified

Epoch

An epoch is simply one complete round of practice for the AI. When the model has looked at absolutely every single piece of data in your training set exactly once, that is one epoch. Running multiple epochs means the model is reviewing the same material multiple times to study and get better, but running too many means it might just memorize the answers (overfitting).

Batch Size

Instead of looking at the entire dataset at once (which requires too much memory) or one by one (which takes forever), the model looks at data in small chunks. The batch size is how many examples the model looks at before it pauses to update what it has learned. A smaller batch size makes the model learn erratically but accurately, while a larger batch size is faster but requires more computing power.

Stratified

This is a method of splitting up your data to keep things fair. If your main dataset is 80% pictures of cars and 20% pictures of bikes, a stratified split ensures that when you divide that data into a "training" pile and a "testing" pile, both piles keep that exact 80/20 ratio. This prevents the model from accidentally getting a training set that is entirely cars.

2. Activation Functions

An activation function is the deciding factor inside a neuron that determines whether the neuron should "fire" or stay quiet based on the information it just received.

Sigmoid

It takes any number and squishes it into a tight range between exactly 0 and 1. It is useful when you need a simple "yes or no" probability, but it is notorious for causing deep networks to stop learning entirely because the outputs get too small to work with.

ReLU (Rectified Linear Unit)

This is the most popular choice for modern neural networks. Its rule is dead simple: if a number is negative, turn it to 0. If it is positive, leave it alone. It makes the network train incredibly fast, but it has a flaw where neurons can "die" and permanently output 0 if they get stuck looking at negative numbers.

Tanh

Similar to Sigmoid, but it squishes numbers between -1 and 1. It centers the data around zero, which usually helps the network learn faster than Sigmoid, but it still suffers from the same issue of getting stuck on extreme numbers.

Softmax

This is the ultimate function for choosing a single category. If you have a model trying to guess if a picture is a dog, cat, or bird, Softmax takes the raw guesses and converts them into clean percentages that add up to exactly 100%.

3. Structured vs. Unstructured Data

Structured Data (Used by ANNs)

This is neat, highly organized data. If you can put it perfectly into an Excel spreadsheet or an SQL database with clear rows and columns (like a customer's age, salary, and location), it is structured. Artificial Neural Networks (ANNs) thrive on this because the inputs are always exactly where they expect them to be.

Unstructured Data (Used by CNNs)

This is messy data that doesn't fit in a spreadsheet: images, audio files, videos, and raw text paragraphs. The information is hidden inside the pixels or soundwaves. Convolutional Neural Networks (CNNs) were specifically built to handle this mess by automatically scanning and finding the patterns themselves.

4. What is a Layer?

A layer is just a line of neurons working together. Data goes in one side, the layer processes it, and spits the new information out to the next layer.

ANN Layers

- **Input Layer:** The front door. It just catches the raw data.

- **Hidden Layer(s):** The brain of the operation. This is where the actual thinking, calculating, and pattern-finding happens.
- **Output Layer:** The final answer. It spits out the prediction.

CNN Layers

- **Convolution Layer:** The scanner. It slides tiny filters over an image to look for basic shapes, like edges, lines, or colors.
- **Pooling Layer:** The compressor. It takes the shapes found by the convolution layer and shrinks them down, throwing away the useless background noise and keeping only the most important features.
- **Fully Connected Layer:** This sits at the very end. It takes all the important features found by the previous layers and makes the final decision (e.g., "Those edges and colors mean this is a picture of a cat").

5. Concepts Used in ANN and CNN

Shared Concepts

- **Weights and Bias:** The actual settings the network learns. Weights decide how important a piece of data is, and bias shifts the answer up or down to make it more accurate.
- **Loss Function:** The scorecard. It calculates exactly how wrong the model's guess was compared to the real answer.
- **Gradient Descent & Backpropagation:** The correction mechanism. When the model gets an answer wrong, these two math processes work backward through the network to tweak the weights so it does better next time.

ANN Specifics

- Relies on fully connected layers, meaning every single neuron talks to every single neuron in the next row.
- Requires the human to figure out what features are important before feeding the data in.

CNN Specifics

- Relies on spatial relationships. It cares about which pixels are next to each other.
- Does its own feature extraction. You just hand it raw images, and it figures out what an "edge" or a "texture" looks like entirely on its own.